Reg. No. | | | | | | | | | | | [12EC72]

SEVENTH SEMESTER B.E EXAMINATION DECEMBER – 2015

MOBILE COMMUNICATION

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	**a.**	Define the parameters required to improve the service quality of cellular communication system.	**04**
	b.	Define cause of Ray leigh fading? State the equation of standing wave pattern in fading.	**04**
	c.	With neat sketch explain basic cellular system and its components.	**06**
2.	**a.**	Explain necessary of frequency reuse concept and types, illustrate with different frequency reuse pattern.	**07**
	b.	Explain handoff mechanism.	**07**
3.	**a.**	Describe the points to be clarified in concern for cell site antenna height effects.	**07**
	b.	Explain Lee's point to point model and its merit.	**07**
4.	**a.**	Explain types of umbrella pattern used in cell site antenna.	**06**
	b.	Give the general equation for both sum and difference for antenna pattern. Discuss how the pattern synthesis problems can be solved from current distribution.	**08**
5.	**a.**	Briefly explain the cases by lowering the height of the antenna may not effectively reduce the interference.	**06**
	b.	Which components of the cellular system control the power levels, and how it will be controlled?	**08**

SCHOOL OF ENGINEERING & TECHNOLOGY LIBRARY DATE JAIN UNIVERSITY BANGALORE

6.	**a.**	List the classification of set up channel.	**02**
	b.	Define concept of channel sharing and channel borrowing.	**04**
	c.	Describe the different types of non-fixed channel assignment algorithms.	**08**
7.	**a.**	State the general formula used for dropped call rate.	**02**
	b.	Explain two-hand off-level algorithm.	**06**
	c.	What are the factors to be taken into consideration for calculating dropped call rate to main certain voice quality level.	**06**
8.	**a.**	Explain the methods to reduce interference in co-channel. Operation.	**07**
	b.	Explain cell splitting techniques, what are the effect on splitting.	**07**